Not All Representations Are Subspaces

A Grassmannian atlas of causal structure in grokking

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Overview

Part 1

What is a causal variable, and where does DAS look?

In this section

1. What is a causal variable?

When we say a neural network “uses” a concept, we mean there is a direction (or subspace) in its activations that, when edited, changes the output in a predictable way.

2. DAS and the Grassmannian

Distributed Alignment Search (DAS) finds these subspaces. Every DAS solution is a point on a specific mathematical object: the Grassmannian manifold. This is the hidden assumption.

3. Why linearity matters

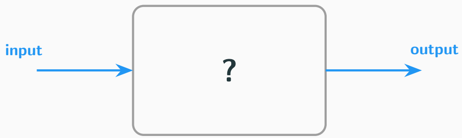
DAS assumes the causal variable is a linear subspace. This works for some variables but fails catastrophically for others — and high IIA scores cannot tell you which case you are in.

1. What is a causal variable?

The interpretability goal (orig)

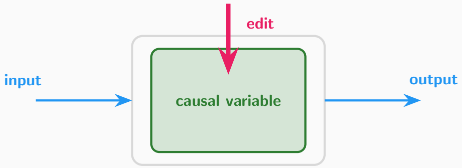
We want to understand *how* neural networks compute, not just *what* they compute.

Standard evaluation



Report accuracy, F1, etc.
Does not explain *why*

Causal abstraction

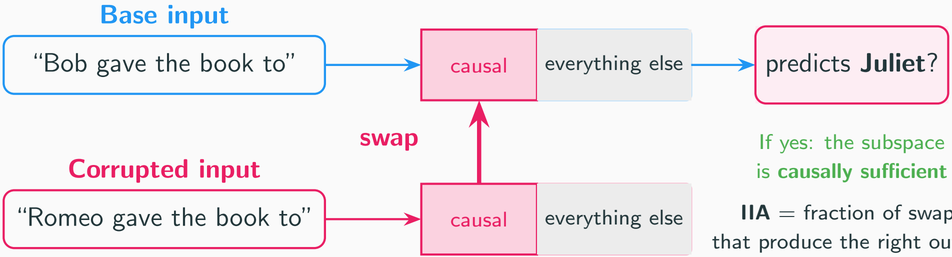


Find internal variables.
Edit them. Check if output changes.

If editing a specific direction in the network's internal representation changes the output predictably, that direction is a **causal variable** for the behavior.

Interchange intervention analysis (IIA) (orig)

The core technique: take two inputs, swap part of their internal representations, check if the output changes accordingly.



If yes: the subspace is **causally sufficient**

IIA = fraction of swaps that produce the right output

2. DAS and the Grassmannian

What DAS actually does (orig)

DAS learns a rotation matrix $Q \in \mathbb{R}^{d \times k}$ that identifies a k-dimensional subspace of the d-dimensional activation space.

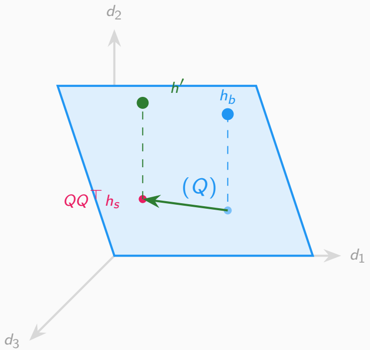
The interchange intervention projects out the base's component and replaces it with the source's:

$$h' = h_b - QQ^\top h_b + QQ^\top h_s$$

This is a **linear** operation. The subspace $QQ^\top$ is an orthogonal projection onto a *flat plane* in activation space.

The set of all k-dimensional planes in $\mathbb{R}^d$ is the **Grassmannian manifold** $\mathrm{Gr}(k, d)$.

Every DAS solution is a point on $\mathrm{Gr}(k, d)$.



DAS projects onto a **flat plane**.

What if the variable lives on a **curve**?

The Grassmannian manifold $\mathrm{Gr}(k, d)$ (orig)

The Grassmannian $\mathrm{Gr}(k, d)$ is the set of all k-dimensional linear subspaces of $\mathbb{R}^d$.

Think of it as a "space of planes." Every DAS solution picks one plane from this space. The distance between two planes is measured by their **principal angles**:

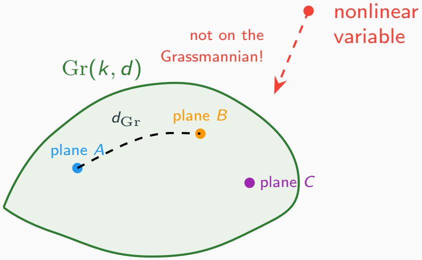
$$d_{\mathrm{Gr}}([Q_1], [Q_2]) = \left(\sum_i \theta_i^2\right)^{1/2}$$

When we ask "does DAS work?" we are really asking:

> Does the true causal variable live on $\mathrm{Gr}(k, d)$?

If yes: DAS finds a meaningful point.

If no: DAS still returns a point, but it is an **artifact**.



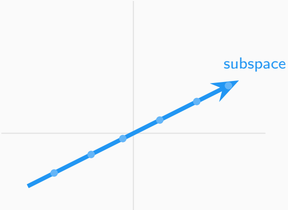
A nonlinear causal variable (circle, sphere)

does not live on $\mathrm{Gr}(k, d)$ at any k.

3. Why linearity matters

Linear vs. nonlinear causal variables

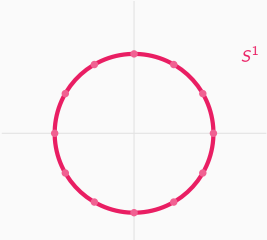
Linear causal variable



Lives on $\mathrm{Gr}(k, d)$ ✓
DAS can find it

Examples: IOI indirect object (partially), multiplication mod p after grokking

Nonlinear causal variable



Not on $\mathrm{Gr}(k, d)$ ×
DAS returns IIA $= 0$

Examples: addition mod p after grokking (Fourier representation encodes numbers on a circle)

The central question

For a given operation: does DAS find a **real** causal variable, or just an **artifact**?

Always

Yes, at k=2, every seed

Stochastic

Depends on the seed

Never

No, regardless of k or seed

We test fourteen modular arithmetic operations and find a **sharp, three-class partition**.

Why subspaces matter for circuits

Mechanistic interpretability asks two questions about neural network circuits:

Q1: Which components?

Which attention heads and MLPs matter for a behavior? Which edges carry the signal?

Methods: activation patching, attribution patching, EAP (Syed et al., 2024).

Q2: What do they communicate?

When two components are connected, *what* flows between them? Which subspaces of the residual stream carry the signal?

Methods: DAS, probing, SAEs.

Q1: *this edge matters*

Q2: *factor* $\mathbf{f_i}$ *carries the signal*

Our work: whether Q2 has a **linear** answer depends on the operation.

Part 2

The atlas: fourteen operations, three classes

4. Fourteen operations, three classes

Experimental setup

We train single-layer transformers on modular arithmetic:

Input: $(a, b, =)$ Output: $f(a, b) \bmod p$

Model: $d_{\text{model}} = 128$, 4 heads, standard grokking setup (30% train, AdamW, weight decay 1.0).

For each trained model, we:

1. Cache residual-stream activations
2. Train DAS at $k \in \{2, 4, 6, \ldots, 32\}$
3. Measure IIA, equivariance, circle geometry
4. Train a structured VAE

14 operations spanning group actions, polynomials, piecewise, and non-algebraic maps.

Group actions (5)

$ab \bmod p$
$(a+b) \bmod p$
$(a-b) \bmod p$
$a/b \bmod p$
$a \oplus b \bmod p$

Polynomials (6)

a^2+b^2, a^3+b^3, a^3
a^2, a^2+b, $2a+3b+5$

Piecewise (2)

$\max(a, b)$, $|a-b|$

Non-algebraic (1)

$a^b \bmod p$

What is grokking?

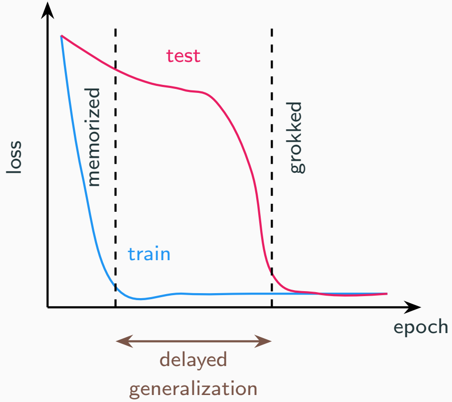
Grokking is a training phenomenon where models memorize the training set long before they generalize.

For modular arithmetic with weight decay:

- Train loss drops to zero quickly (memorization)
- Test loss stays high for thousands of epochs
- Then suddenly drops (generalization)

The internal representation changes during grokking: from a lookup table (memorization) to a structured algorithm (Fourier representation).

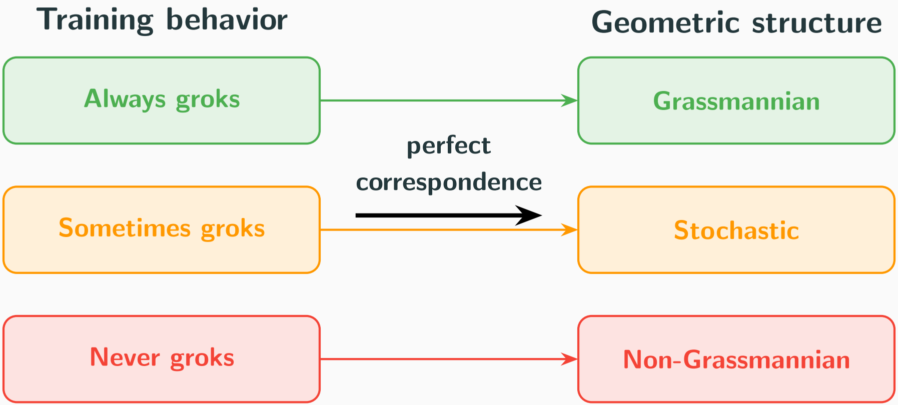
Prior work showed grokked models use Fourier features:
$a \mapsto (\cos 2\pi fa/p, \sin 2\pi fa/p)$



The atlas: three sharp classes

Operation	Grok	Loss	IIA(k=2)	k^*	Equiv.	Class
Multiplication	✓	0.000	1.000	2	0.995	Always
Subtraction	✓	0.000	1.000	2	1.000	Always
Division	✓	0.000	1.000	2	1.000	Always
Bitwise XOR	✓	0.003	1.000	2	1.000	Always
Cubic sum	✓	0.000	1.000	2	1.000	Always
Sum of squares	✓	0.005	1.000	2	0.987	Always
Max	✓	0.074	0.960	2	0.957	Always
Power (s137)	✓	0.088	0.940	4	0.965	Stochastic
Power (s42)	×	1.143	0.980	2	0.665	Stochastic
Comp. add. (orig)	✓	0.002	1.000	2	0.998	Stochastic
Comp. add. (s42)	×	3.412	0.860	>32	0.110	Stochastic
Cubing	×	9.765	0.857	4	0.509	Never
Squaring	×	7.808	0.857	4	0.314	Never
Abs. difference	×	1.835	0.800	>32	0.189	Never
Polynomial	×	21.55	0.720	>32	0.015	Never
Affine	×	26.21	0.780	>32	0.026	Never

The finding: grokking predicts geometry



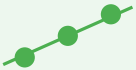
Grokking is the mechanism: when generalization happens, the representation becomes a linear subspace on $\mathrm{Gr}(k, d)$. When it doesn't, the variable stays nonlinear.

Three classes, one mechanism

Always Grassmannian

7 operations

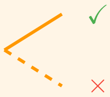
- All grokked
- IIA $= 1.0$ at $k = 2$
- Equivariance $> 95\%$
- Causal variable is a linear subspace



Stochastic

2 operations (4 seeds)

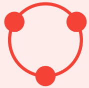
- Same operation
- Same hyperparameters
- Different random seed
- Opposite outcomes!
- Grokking is the switch



Never Grassmannian

5 operations

- None grokked
- IIA high but via lookup
- Equivariance $< 52\%$
- Causal variable is nonlinear or absent



5. Linear DAS fails completely

Linear DAS returns zero IIA on grokked modular addition

Addition mod 113 groks perfectly (test loss $< 10^{-7}$). The model has learned the correct algorithm.

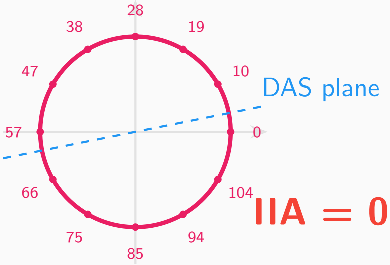
But DAS returns $\mathrm{IIA} = 0.0$ at **every** subspace dimension $k \in \{2, 4, 8, 16\}$.

This is not an optimization failure. The model encodes numbers using Fourier features:

$$a \mapsto \left(\cos \tfrac{2\pi f a}{p},\ \sin \tfrac{2\pi f a}{p}\right)$$

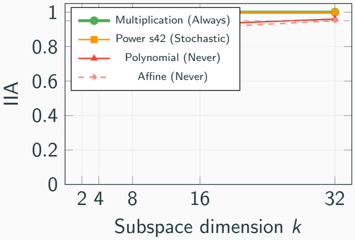
This representation lives on a **circle** S^1 in $\mathbb{R}^d$, not a linear subspace.

DAS searches $\mathrm{Gr}(k, d)$ for a flat plane that captures a curved manifold — a **category error**, not an optimization failure.



Numbers on S^1

k-sweep profiles distinguish the three classes



Always Grassmannian: IIA saturates at 1.0 by $k = 2$. The causal variable is intrinsically 2-dimensional.

Never Grassmannian: IIA rises gradually and may not reach 1.0 even at $k = 32$. More dimensions help, but the fit is never clean.

The k-sweep profile is a **diagnostic**: if IIA keeps rising past $k = 4$, the causal variable is probably not Grassmannian.

6. Stochastic grokking

Stochastic grokking: same operation, opposite outcomes

The stochastic class is the **strongest evidence**:

Power (a^b mod 113):

- Seed 137: grokked. Equiv = 96.5%
- Seed 42: not grokked. Equiv = 66.5%

Composite addition ($(a + b)$ mod 91):

- Original seed: grokked. Equiv = 99.8%
- Seed 42: not grokked. Equiv = 11.0%

Same operation. Same architecture. Same hyperparameters. **Opposite outcomes from random initialization alone.**

Grassmannian variables appear if and only if the model generalizes.

Power: seed 137

Grokked
Loss = 0.088
Equiv = 96.5%

Same operation
Same architecture
Same hyperparams
Different seed

$\neq$

Power: seed 42

Not grokked
Loss = 1.143
Equiv = 66.5%

Equivariance: the geometric diagnostic

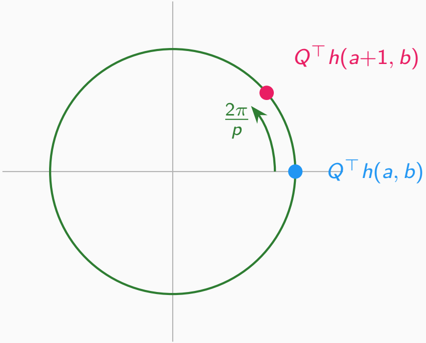
For operations with a group action, the DAS subspace should respect the algebraic symmetry.

Test: shift input $a \to a + 1 \bmod p$. Does the DAS projection rotate by $\mathbf{2\pi/p}$?

Random subspaces: equivariance $< 1\%$.
Always Grassmannian: equivariance $> 95\%$.
Never Grassmannian: equivariance $< 52\%$.

Equivariance distinguishes:

- Genuine structure from memorization
- Causal variables from artifacts
- What IIA alone cannot tell apart



If the variable is Grassmannian, shifting input = rotating projection

Part 3

The vacuity problem: when high IIA means nothing

7. Memorization artifacts

Memorization produces high IIA

Squaring (a^2 mod 113) achieves $\mathrm{IIA} = 0.86$ at $k = 2$ and $\mathrm{IIA} = 1.0$ by $k = 4$.

But the model **never grokked** (test loss = 7.8). It memorized the training set.

Equivariance exposes the artifact:

- Squaring: 31.4% (vs. $> 95\%$ for genuine)
- Cubing: 50.9%
- Random subspace: $< 1\%$

High IIA with low-k saturation is compatible with a **lookup table**. IIA alone cannot tell the difference.

Genuine (multiplication)

IIA = 1.0 at $k = 2$
Equiv = 99.5%

✓ **Structured causal variable**

IIA alone cannot tell these apart!

Artifact (squaring)

IIA = 1.0 at $k = 4$
Equiv = 31.4%

× **Lookup table**

8. Nonlinear DAS is vacuous

The natural response: add nonlinearity

DAS assumes linearity and fails on nonlinear variables. Natural fix: prepend an MLP featurizer.

NL-DAS: learn encoder f and decoder g, then apply linear DAS in the transformed space:

$$h' = g\big(f(h_b) - QQ^\top f(h_b) + QQ^\top f(h_s)\big)$$

Result on IOI (GPT-2):

- DAS at $k = 1$: IIA $= 0.193$
- NL-DAS at $k = 1$: IIA $= \mathbf{1.000}$

This looks like a dramatic improvement.

It is not.

DAS (linear)

$$\mathbf{h} \xrightarrow{\mathbf{QQ}^\top} \mathbf{h}'$$

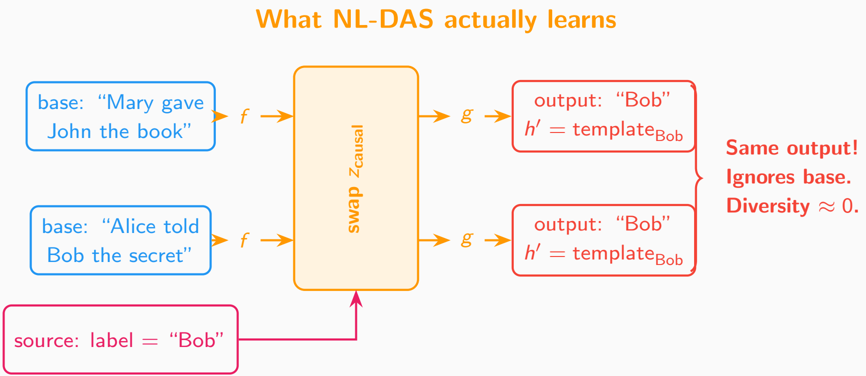
NL-DAS (nonlinear)

$$\mathbf{h} \xrightarrow{\mathbf{f}} \mathbf{z} \xrightarrow{\mathbf{QQ}^\top} \mathbf{z}' \xrightarrow{\mathbf{g}} \mathbf{h}'$$

f and g trained only on interchange loss

No reconstruction constraint!
$\mathbf{g} \circ \mathbf{f} \neq$ identity

The lookup-table failure mode



The decoder produces a **canonical activation for each class**, regardless of base input.

This achieves IIA $= 1.0$ but is a **lookup table**, not a causal variable.

NL-DAS vs. DAS vs. structured VAE

Task	k	DAS	NL-DAS	VAE	Assessment
IOI	1	0.193	**1.000**	0.232	NL-DAS vacuous
IOI	2	0.297	**1.000**	0.719	NL-DAS vacuous
IOI	4	0.556	**1.000**	**0.978**	VAE genuine
Addition	1	0.006	0.194	0.000	Both honest
Addition	4	0.289	**0.883**	0.089	NL-DAS suspect
Mult.	4	0.344	**0.961**	0.000	NL-DAS suspect
Squaring	$*$	0.000	0.011	0.000	No structure

Red bold: suspiciously high IIA. Green bold: validated by diversity ratio.

NL-DAS achieves IIA ≈ 1 on IOI at $k = 1$ — that is getting perfect interchange accuracy from a **single dimension**. Diversity ratio ≈ 0 confirms it is a lookup table.

9. Intervention faithfulness

The diversity ratio: detecting lookup tables

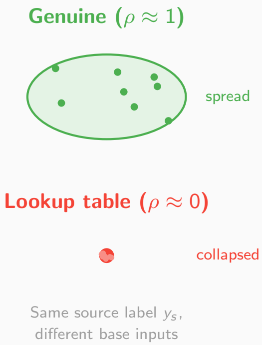
For each source label y_s, compute the standard deviation of intervened activations h' across all base inputs:

$$\rho = \frac{\mathbb{E}_{y_s}\left[\mathrm{std}(h'_{y_s})\right]}{\mathbb{E}_{y_s}\left[\mathrm{std}(h_{y_s})\right]}$$

$\rho \approx 0$: all intervened activations for the same source label collapse to a single point. **Lookup table.**

$\rho \approx 1$: the intervention preserves natural variation within each class. **Genuine intervention.**

This is the key metric that exposes NL-DAS: it achieves IIA $= 1.0$ but $\rho \approx 0$.



Four complementary faithfulness metrics

Diversity ratio ρ

Does the intervention preserve variation within each class?

$\rho \approx 0$: lookup table

$\rho \approx 1$: genuine

Reconstruction MSE

Can the decoder reconstruct the original activation?

High MSE + high IIA: decoder is "hallucinating"

KL / JS divergence

How much does the full output distribution change?

Lower = more faithful.

Binary IIA hides this.

Normalized logit diff

Does the intervention reproduce the clean model's confidence?

1.0 = exact match

$\gg 1$ or $\ll 0$ = distortion

Why this matters

Binary IIA treats a 51% probability shift the same as a 99% shift.

These continuous metrics distinguish among methods that all achieve IIA ≈ 1.0.

Report at least one faithfulness metric alongside IIA.

Part 4

The fix: structured VAE and cross-task transfer

10. Structured VAE

Structured VAE for nonlinear causal variables

The structured VAE partitions the latent space into:

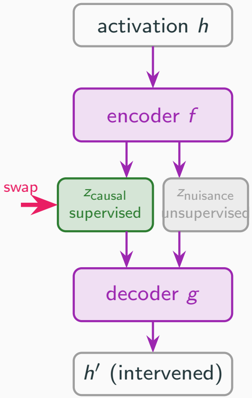
- z_{causal}: supervised with the operation's label
- z_{nuisance}: unsupervised, absorbs everything else

The loss combines three terms:

$$\mathcal{L} = \underbrace{-\mathbb{E}_q[\log p(x \mid y, z)]}_{\text{reconstruction}} + \beta \underbrace{D_{\text{KL}}}_{\text{regularization}} + \alpha \underbrace{\mathcal{L}_{\text{CE}}}_{\text{supervision}}$$

The ELBO prevents the lookup-table failure:

1. Reconstruction forces faithful decoding
2. KL prevents latent collapse
3. Causal/nuisance split preserves base info



Why the VAE is not vacuous: three constraints

NL-DAS (vacuous)

- × No reconstruction loss
- × No KL regularization
- × No causal/nuisance split
- × Trained only on IIA
- → Can learn lookup table
- → Diversity $\rho \approx 0$

vs.

Structured VAE (genuine)

- ✓ Reconstruction → faithful decoding
- ✓ KL → prevents latent collapse
- ✓ Causal/nuisance split
- ✓ ELBO + supervision
- → Cannot learn lookup table
- → Diversity $\rho \approx 1$

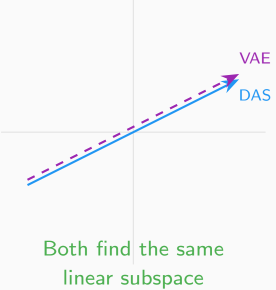
Controls confirm the VAE is not cheating

Random-label VAE: IIA ≤ 0.02 | Reconstruction-only ($\alpha = 0$): IIA ≤ 0.005 | Untrained: IIA $= 0.000$
Plain VAE (no causal/nuisance split): IIA ≤ 0.08 — Correct supervision AND structure are both required

VAE matches DAS where DAS works, extends where it fails

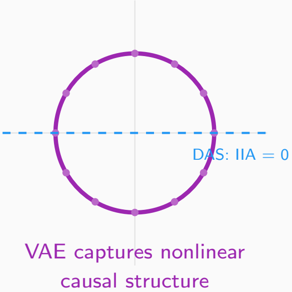
Always Grassmannian operations:

VAE reduces to an effective rotation.



Never Grassmannian operations:

VAE's nonlinear encoder captures curved structure.



For grokked addition and multiplication, both DAS and VAE achieve IIA $= 1.000$ and equivariance $> 99\%$.

The VAE is a **strict generalization** of DAS: when the variable is linear, VAE $=$ DAS.

11. Cross-task transfer

Cross-task transfer validates causal variables

A genuine causal variable should **transfer** — train on one task variant, test on another.

IOI cross-template transfer (GPT-2):

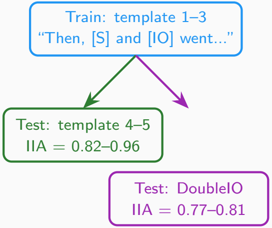
- Train VAE on templates 1–3
- Test on unseen templates 4–5
- IIA = 0.82–0.96 at $k = 4$

Jensen's extended IOI tasks:

- DoubleIO: IIA = 0.77–0.81
- TripleIO: IIA = 0.77–0.81
- Trained only on base IOI!

IOI subtask transfer (8 MIB counterfactual types):

Transfer validation



A lookup table cannot transfer like this

12. IOI is not Grassmannian

IOI is NOT Grassmannian: sheaf consistency analysis

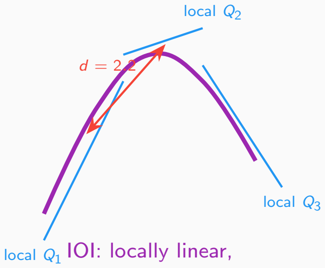
Even within GPT-2's IOI circuit, the causal variable is not globally linear.

We fit local DAS subspaces to different clusters of activations and measure consistency:

- Geodesic distance between local subspaces: $\sim$**2.2**
- Cross-cluster IIA: $\sim$**0.21**

If IOI were Grassmannian, local subspaces would agree (geodesic ≈ 0, cross-IIA ≈ 1.0).

The subspace that represents the indirect object **rotates** across different activation clusters. Locally linear, globally nonlinear — like a curved manifold approximated by tangent planes.



From subspaces to interpretable factors

DAS finds a subspace. But which *directions* compose it?

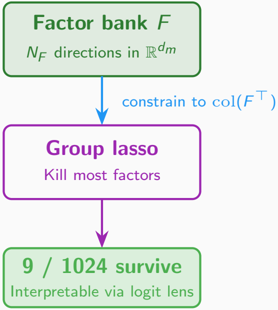
If the model's weight matrices decompose as $W_i = S_i \cdot F$ through a shared **factor bank** F with sparse **selectors** S_i, then DAS can be constrained to search within the factor span:

Standard DAS: $U = \mathrm{orth}(A),\ A \in \mathbb{R}^{d_m \times k}$

Factorized DAS: $U = \mathrm{orth}(F^\top A),\ A \in \mathbb{R}^{N_F \times k}$

With a group lasso penalty ($\lambda \sum_i \|A_{i,:}\|_2$), most factors are killed. What survives is interpretable:

Task	Active	Total	Top factor decodes to
IOI	9	1024	*wives, women, Ms*
Gender bias	3	1024	*she, her, herself*
SVA	2	1024	*are, is, aren't*



Standard DAS finds the subspace. Factorized DAS finds the **factors** in it.

Weight space vs. activation space

Both SAEs and sparse weight factorization produce dictionaries of directions in the residual stream. But they decompose different things:

SAEs (activation space)

frozen model

↓

SAE (post-hoc)

Decomposes what the model *outputs* at a layer.

Not guaranteed to capture what the model actually *uses*.

Factors (weight space)

$W = S \cdot F$

↓

computation

Decomposes how the model *computes*: $W_i = S_i \cdot F$.

Guaranteed to be used — every active factor is part of the computation.

The Grassmannian question applies to both: whether the causal variable is linear determines whether SAE features, DAS subspaces, or factor-based

Part 5

Implications and practical advice

13. Four-level hierarchy

A four-level hierarchy for causal validity

Level 1: Interchange effectiveness (IIA)

Necessary, not sufficient. Squaring gets IIA $= 1.0$ without structure.

Level 2: Interchange faithfulness (ρ, recon)

Kills NL-DAS. Automatic for linear DAS. Required for nonlinear.

Level 3: Distributional fidelity (KL, JS, logit diff)

Distinguishes quality among methods with IIA ≈ 1.0.

Level 4: Structural diagnostics (equiv, transfer)

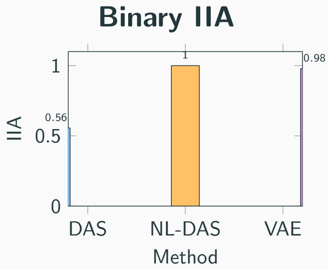
Strongest evidence. Equivariance or cross-distribution generalization.

stronger evidence

Key asymmetry: Linear DAS automatically passes Level 2 (orthogonal projection is faithful by construction) but may fail Level 1 on nonlinear variables. NL-DAS easily passes Level 1 but fails Level 2.

14. Continuous metrics

Binary IIA hides important differences



NL-DAS "wins." But it is a lookup table.

With faithfulness metrics

	DAS	**NL-DAS**	**VAE**
IIA	0.556	1.000	0.978
Diversity ρ	1.0	≈ 0	≈ 1
Transfer	Yes	No	Yes

VAE wins genuinely. NL-DAS is disqualified by $\rho \approx 0$.

DAS is honest but limited by linearity.

15. Practical advice

Practical advice for causal abstraction

1. **Never report IIA alone.** IIA $= 1.0$ is compatible with genuine structure, memorized lookup tables, and degenerate encoder-decoders. Always report at least one faithfulness metric.
2. **Use continuous distributional metrics.** KL divergence and normalized logit difference distinguish among methods that all achieve IIA ≈ 1.0. Binary IIA treats a 51% shift the same as 99%.
3. **Be suspicious of unconstrained nonlinear featurizers.** Any encoder-decoder trained end-to-end on interchange loss without reconstruction constraints can learn a lookup table. The ELBO (or equivalent) is necessary.
4. **Test cross-distribution generalization.** Evaluate on disjoint names, novel templates, or new task variants. A method that fails under distribution shift is exploiting surface regularities.
5. **Consider nonlinear methods when DAS IIA is low.** Low IIA does not mean no causal

Surprising positive and negative cases

Surprising positives

Cubic sum ($a^3 + b^3$): perfect equivariance (100%) despite cubing alone reaching only 50.9%. The boundary is not "group action vs. polynomial" but whether the binary structure provides sufficient additive symmetry.

Max ($\max(a, b)$): achieves 95.7% equivariance despite not being a group action. It is piecewise equivariant: when the argmax is stable under shift, max behaves like a shifted identity.

Surprising negatives

Affine ($2a + 3b + 5$): a *linear* function that *fails* to produce a Grassmannian variable! Under $a \mapsto a + g$, the output shifts by $2g$, not g. Equivariance requires the operation to be a group action, not merely linear.

Squaring: achieves IIA $= 1.0$ by $k = 4$ without grokking. High IIA is the illusion; low equivariance (31.4%) is the truth.

16. Conclusion

Summary

1. 14 operations $\to$ three sharp classes (Always / Stochastic / Never Grassmannian). Boundary governed by grokking.

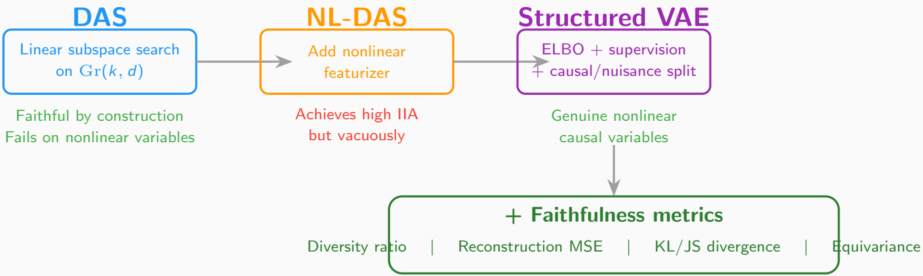
2. Linear DAS returns IIA $= 0$ on grokked addition. The variable lives on S^1, not $\mathrm{Gr}(k, d)$. Category error.

3. Unconstrained NL-DAS achieves IIA $= 1.0$ vacuously (lookup table). Diversity ratio ρ exposes this.

4. Structured VAE recovers genuine nonlinear causal variables. ELBO prevents degeneracy. Transfers across tasks.

The linearity assumption is **testable**. The natural fix (nonlinearity) introduces a new failure mode. This paper provides the diagnostics to detect both problems and the method to avoid them.

The big picture



Every causal abstraction claim should be evaluated with the **four-level hierarchy**.

IIA is necessary but never sufficient.

Thank you

Code and experiments:

`github.com/elliottower/grokking-causal-geometry`

Elliot Tower · Ivan Vegner · N. Siddharth · Alex Doumas

Draft paper: *When Does Linear Causal Abstraction Work? Mapping the Boundary on the Grassmannian*